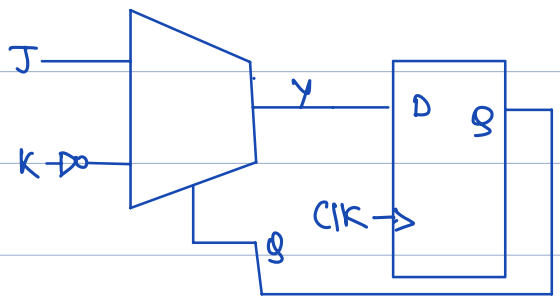
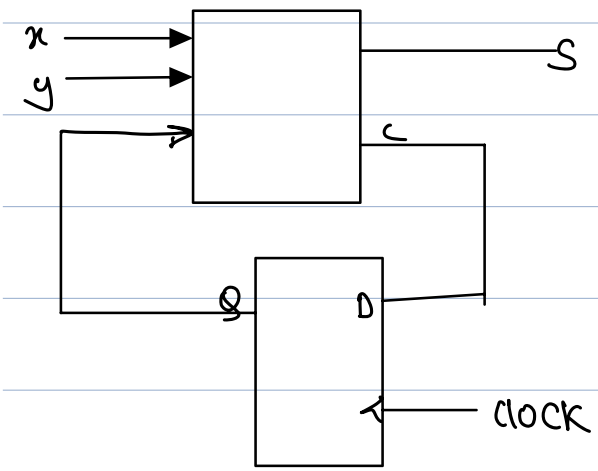
Construct a JK flip-flop using a D-flipflop; two one-to-one line multiplexer, and an inverter

$Y = Q'J + QK'$



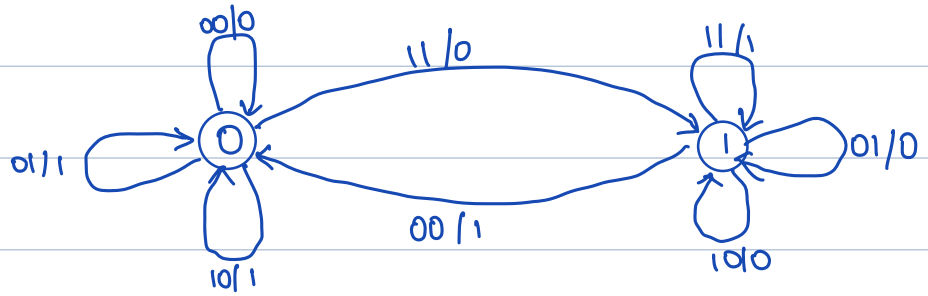


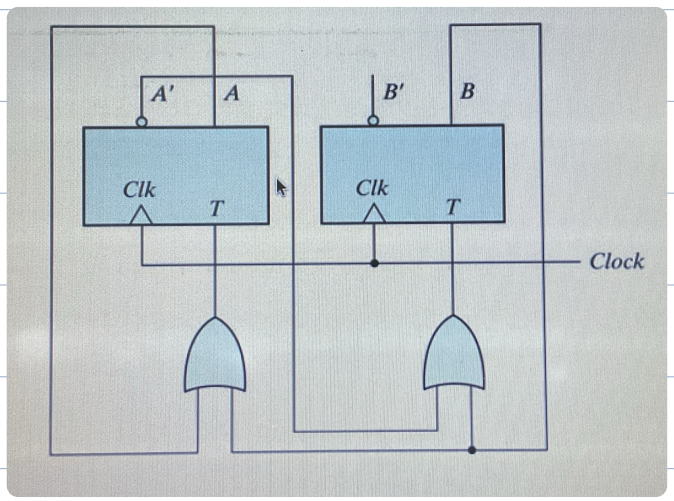
write state table and draw state diagram

$C = D = Q(t+1) = xy + yQ(t) + xQ(t)$

$S = x \oplus y \oplus Q(t)$

present	x	y	next	output
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1





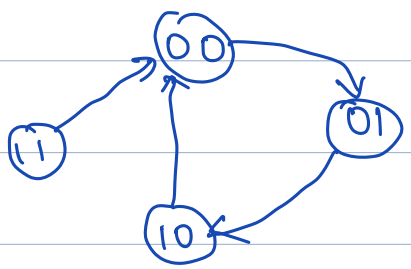
Draw state table and state diagram, Explain the function this circuit performs

$T_A = Q_A + Q_B \qquad T_B = Q_B + \overline{Q_A}$

$Q_A^+ = Q_A \oplus (Q_A + Q_B) = \overline{Q_A} Q_B$

$Q_B^+ = Q_B \oplus (Q_B + \overline{Q_A}) = \overline{Q_B}\,\overline{Q_A}$

Q_A	Q_B	Q_A^+	Q_B^+
0	0	0	1
0	1	1	0
1	0	0	0
1	1	0	0



$00 \rightarrow 01 \rightarrow 10$

mod 3 counter